than a probability distribution so structureless that every possible outcome is equally probable?

Maximal entanglement implies a complete lack of information about Alice's subsystem for experiments that only involve that one subsystem. On the other hand, it implies a large correlation between Alice's and Bob's measurements. For the singlet state, if Alice measures any component of her spin, she automatically knows the result Bob would get if he were to measure the same component of his spin. This is exactly the kind of knowledge that is precluded in a product state.

So in each type of state, some things are predictable and some are not. In a product state, we can make statistical predictions about measurements made on each separate subsystem, but Alice's measurements tell her nothing about Bob's system. In a maximally entangled state, on the other hand, Alice can predict nothing about her own measurements, but she knows a great deal about the relation between her outcomes and Bob's.

7.8 THE PROCESS OF MEASUREMENT

We have seen that quantum systems evolve in what look like irreconcilably different ways: by unitary evolution between measurements, and by wave function collapse when measurements take place. This circumstance has led to some of the most contentious debates and confusing claims about so-called reality. I'm going to steer away from those debates and stick to the facts. Once you know how quantum mechanics works, you can decide for yourself whether you think there is a problem.

Let's begin by noting that every measurement involves a system and an apparatus. But if quantum mechanics is a consistent theory, then it should be possible to combine the system and apparatus into a single bigger system. For simplicity let's take the system to be a single spin. The apparatus $\mathcal{A}$ is the same one that we used in the very first lecture. The window in the apparatus can show three possible readings. The first is blank—it represents the neutral state of the apparatus before it comes in contact with the spin.